Darshan Singh S

Pre-Doctoral Researcher, Google DeepMind

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Education

International Institute of Information Technology Hyderabad (IIIT-H) **2022–2024**
M.S by Research in Computer Science CGPA: **9.67/10.0**

Experience

Google DeepMind **Nov 2024 - Present**
Pre-Doctoral Researcher
Projects: Gemini Core team member - multimodal-multicultural video understanding and reasoning, multimodal pretraining and posttraining evals, knowledge probing, cross-lingual gaps analysis in language models.
Languages Team: I primarily work with Partha Talukdar, Shachi Dave, Arsha Nagrani and Tobias Weyand

FastCode AI **June 2024 - Nov 2024**
Machine Learning Engineer
Projects: Developed a time-series foundation model for the electric vehicles data for Mercedes-Benz Research & Development, fine-tuned stable diffusion models and adapted them to video games.
Team: Dr. Arjun Jain, Mercedes-Benz Research & Development, India

CVIT Lab, IIIT Hyderabad **July 2021 - Aug 2022**
Research Fellow | Advisors: Prof. C. V. Jawahar and Prof. Makarand Tapaswi
Projects: Audio-Visual lecture segmentation, published at WACV 2023.
Team: Video-Language (Multimodal) team

Mercedes Benz R&D India **Aug 2019 - Aug 2020**
Graduate Engineer Trainee
Projects: Implemented graph algorithms for enhancing functionalities of a CAD software platform called Micro-Station.

Publications

†=Equal Contribution

[1] **VELOCITI: Benchmarking Video-Language Compositional Reasoning with Strict Entailment** 🤗
Darshana S†, Varun Gupta†, **Darshan Singh S**†, Zeeshan Khan, Vineeth Gandhi, Makarand Tapaswi.
Computer Vision and Pattern Recognition (CVPR). **[CVPR'25]**

[2] **CURVE: A Benchmark for Cultural and Multilingual Long Video Reasoning.**
Darshan Singh, Arsha Nagrani, Kawshik Manikantan, Harman Singh, Dinesh Tewari, Tobias Weyand, Cordelia Schmid, Anelia Angelova, Shachi Dave. *Under Review, 2025.*

[3] **SRL-CLIP: Efficient CLIP Video Adaptation via Structured Semantic Role Labels**
Darshan Singh S, Zeeshan Khan, Makarand Tapaswi. *Under Review, 2025.*

[4] **Unsupervised Audio-Visual Lecture Segmentation**
Darshan Singh S†, Anchit Gupta†, C. V. Jawahar, Makarand Tapaswi.
Winter Conference on Applications of Computer Vision (WACV). **[WACV'23]**

[5] **No Detail Left Behind: Revisiting Self-Retrieval for Fine-Grained Image Captioning**
Manu Gaur, **Darshan Singh S**, Makarand Tapaswi.
Transactions of Machine Learning Research (TMLR). **[TMLR'24]**
Top 10% Distinction, to be presented at ICLR 2026.

[6] **D3: Evaluating MLLMs capacity for Finegrained Visual Discrimination through Self-Retrieval**
Manu Gaur, **Darshan Singh S**, Makarand Tapaswi.
ECCV-Eval-FoMo Workshop. **[ECCV-W'24]**

[7] **Rethinking Cross-lingual Gaps from a Statistical Viewpoint**
Vihari Piratla, Purvam Jain, **Darshan Singh S**, Trevor Cohn, Partha Talukdar. *Under Review, 2025.*

Selected Projects

Gemini (Core-team member) Google DeepMind
Team: Partha Talukdar, Arsha Nagrani, Shachi Dave, Tobias Weyand, Anelia Angelova, Cordelia Schmid

- Developed multimodal-multicultural video understanding and reasoning benchmarks which are used in pre-training and post-training evals.
- Led an effort on reliable knowledge estimation in LLMs and contributed to quantifying cross-lingual gaps in LLMs.

FiGCLIP: Fine-Grained CLIP Adaptation via Densely Annotated Videos IIIT Hyderabad
Advisors: Prof. Makarand Tapaswi

- Addressed the lack of fine-grained and syntactic information in CLIPs representations by adapting CLIP on holistic, multidimensional, and densely annotated video-text data.
- Proposed a lightweight adaptation strategy with LoRA adapters that enable learning fine-grained representations without catastrophic forgetting.
- Outperformed the base CLIP across 5 diverse tasks including video situation recognition, zero-shot action recognition, dense video captioning, and VL compositional reasoning.

VELOCITI: Benchmarking Video-Language Compositional Reasoning IIIT Hyderabad
Advisors: Prof. Makarand Tapaswi, Vineeth Gandhi

- Proposed VELOCITI, a new benchmark to evaluate how well Video-LLMs understand compositional reasoning, and introduced StrictVLE, a stricter evaluation method.
- Key finding: State-of-the-art models like Gemini 1.5 Pro (49.3%) perform far below human accuracy (93.0%), showing significant struggles in associating agents with actions.
- Provided key insights, such as understanding actions is harder than agents for open models and smaller models are predisposed towards Yes for the entailment task.

No Detail Left Behind: Revisiting Self-Retrieval for Fine-Grained Image Captioning
Advisors: Prof. Makarand Tapaswi

- We design- (1) a post-training recipe for self-retrieval finetuning with REINFORCE, and (2) a synthetic framework for visually boosting captioning datasets
- Jointly, they enable captioners to generate fine-grained, succinct descriptions while reducing hallucinations.
- Using our training recipe, ClipCap, a 200M param simplication of modern MLLMs, outperforms sota open-source MLLMs on fine-grained visual discrimination.

Unsupervised Audio-Visual Lecture Segmentation IIIT Hyderabad
Advisors: Prof. C. V. Jawahar, Prof. Makarand Tapaswi

- Proposed a method to segment video lectures into bite-sized topics using visual, textual, and OCR cues via a self-supervised pretext task.
- Utilized learned representations to temporally segment lectures using the TW-FINCH algorithm.
- Introduced AVLectures, a new large-scale dataset of 86 courses (2,350+ lectures) from MIT-OpenCourseWare for pre-training, fine-tuning, and evaluation.

Technical Skills

Languages	Python, C, C++, MATLAB, Assembly Language(8051), Embedded C
Tools/Libraries	PyTorch, NumPy, scikit-learn, LaTeX, Git, HuggingFace

Relevant Coursework

Statistical Methods in AI*, Topics in Deep Learning*, Advanced NLP, Computer Vision, Data Structures and Algorithms*, Digital Image Processing*, Fairness, Privacy & Ethics in AI*, Operating Systems, Programming in C*, Advanced Calculus*, Multivariable Calculus*, Probability and Statistics.
(indicates grade A for outstanding performance)*

Selected Honors, Awards and Academic Service

- Awarded J2C Certification (top 10%) for our paper at TMLR 2024
- Selected for competitive Pre-doctoral position at Google DeepMind
- Presented my work on reliable knowledge estimation at Google DeepMind APAC forum and Multilinguality Research Forum

- Gave a talk on Computer Vision Fundamentals along with Pytorch tutorials at the CVIT Computer Vision Summer School, IIIT Hyderabad
- Started Vision Reading Group at IIIT Hyderabad
- Reviewer: ECCV 2024 (including emergency reviewing), ACCV 2025, CVPR 2025 (including emergency reviewing)

References

References from the following available upon request:

- Partha Talukdar .... Principal Scientist, Google DeepMind []
- Arsha Nagrani .... Senior Research Scientist, Google DeepMind []
- Anelia Angelova .... Principal Scientist, Google DeepMind []
- Shachi Dave .... Research Engineer, Google DeepMind []
- Prof. C.V. Jawahar .... Professor, Dean R&D, IIIT Hyderabad []
- Prof. Makarand Tapaswi .... Assistant Professor, IIIT Hyderabad []
- Prof. Vineet Gandhi .... Associate Professor, IIIT Hyderabad []